# ArthVruksh — FQS Model

# Walk-Forward, Point-in-Time Backtest Report

Goal-based validation: recommendations frozen at each historical decision date, graded only against NAV data that arrived after that date.

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**Headline result (pooled across 100 monthly decision dates, Apr-2013 to Jul-2021): Funds the model ranked in its top decile at the decision date went on to realise a mean 5-year Wealth Multiplier of 1.52x vs 1.42x for the full eligible universe — a +6.8% uplift — and hit their category's goal-hurdle WM 60.3% of the time vs 52.0% for the universe average.**

Prepared for ArthVruksh · Data: 164 equity & hybrid schemes, month-end NAVs 2006–2026 · Report generated 10-Aug-2026

# 1. Methodology — Why This Is Not a Look-Ahead Backtest

Every result in this report follows a strict two-clock protocol, matching institutional walk-forward practice. At each historical **decision date** (month-end, Apr-2013 through Jul-2021), the model computes FQS / blended Wealth-Multiplier score using **only** NAV history available up to that date — every 5-year rolling window used to build the score closes on or before the decision date, never after it. Funds are ranked and the recommendation is frozen. Only after freezing does the report "unlock" the NAVs that actually arrived over the following 5 years, purely to grade whether the frozen call was good. The model never sees those future NAVs when making the call.

This mirrors the existing point-in-time engine already built into the codebase (`backtest/point_in_time.py`, `backtest/backtest_engine.py`), which truncates the NAV store to `date ≤ cutoff` before any scoring happens, and re-derives eligibility and score from scratch at every cutoff. This report is built directly on that engine's full production run (100 monthly cutoffs, completed 07-Aug-2026) — it is not a re-implementation.

## Setup used in this backtest

| | |
|---|---|
| **Universe** | 164 equity & hybrid schemes (Large/Mid/Small/Multi/Flexi Cap, ELSS, Hybrid) — Debt and Gold/Silver exclu |
| **Decision dates** | 100 month-end cutoffs, 30-Apr-2013 → 31-Jul-2021 (every cutoff whose 5-year forward window has fully rea |
| **Ranking metric** | blended_wm — the app's primary production ranking metric (FQS is the headline score shown to users; both |
| **Selection tested** | Top decile (top 10%) of eligible funds by rank, per cutoff — the funds ArthVruksh would actually have surfac |
| **Holding horizon** | 5-year forward SIP (■10,000/month), realised using actual NAVs Jan of decision-year+1 through the 5-year |
| **Goal-achievement bar** | Category-specific target Wealth Multiplier ("goal-hurdle WM") from the FQS category table — e.g. Small Cap |
| **Observations** | 13,574 validated fund × decision-date predictions with fully realised 5-year outcomes |

Note on the goal-hurdle: this is ArthVruksh's internal category target used to score funds, not a market index (e.g. Nifty). A true market-index overlay is not yet wired into the backtest data and is flagged as a follow-up in Section 5.

## 2. Decision-by-Decision: Would the Top-FQS Picks Have Worked?

For each 31-December decision date from 2013 to 2020, the chart below freezes the top-decile FQS/blended-WM picks and the full eligible universe, then shows what each group actually realised over the following 5 years (chart uses realised NAVs only, never seen by the model at decision time).

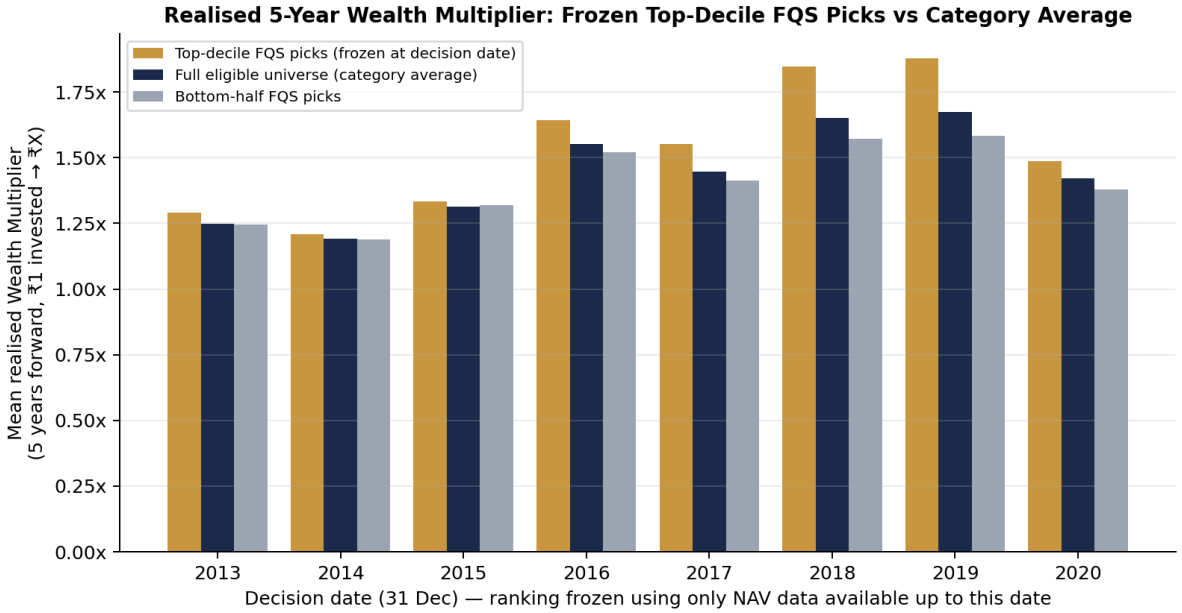



Fig. 1 — Mean realised 5-year Wealth Multiplier, by decision year and cohort.

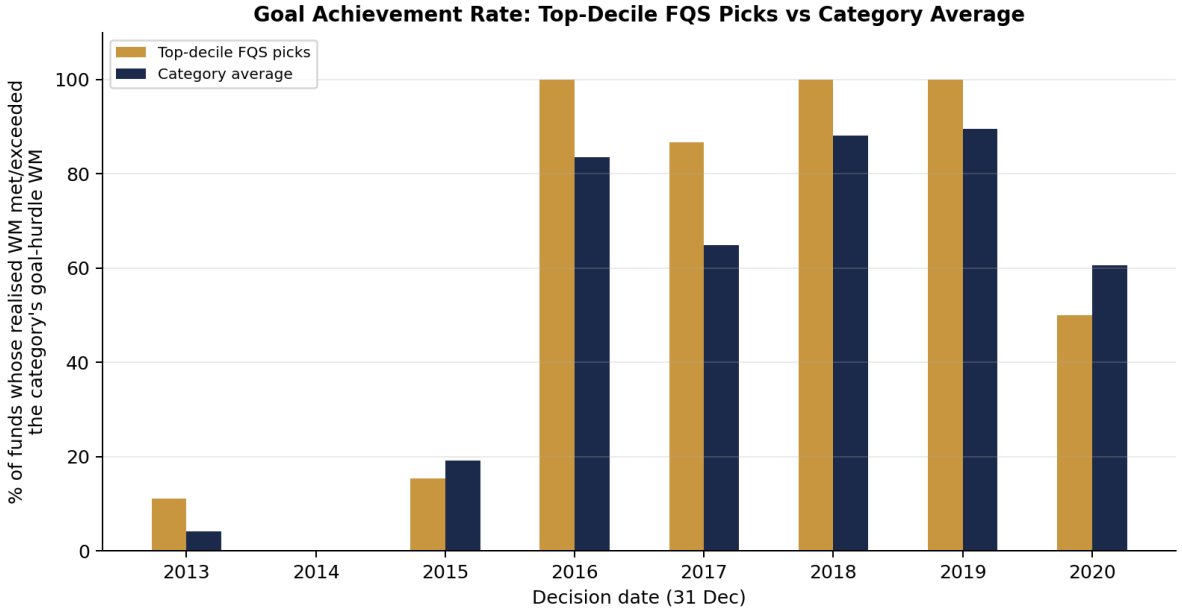



Fig. 2 — Share of funds whose realised outcome met or exceeded the category's goal-hurdle WM.

### Year-by-year detail

| Decision date | Eligible funds | Top-decile mean WM | Category mean WM | Top-decile goal-hit % | Category goal-hit % |
|---|---|---|---|---|---|
| 2013 | 97 | 1.29x | 1.25x | 11% | 4% |
| 2014 | 117 | 1.21x | 1.19x | 0% | 0% |
| 2015 | 131 | 1.33x | 1.31x | 15% | 19% |

| | | | | | |
|---|---|---|---|---|---|
| 2016 | 139 | 1.64x | 1.55x | 100% | 83% |
| 2017 | 148 | 1.55x | 1.45x | 87% | 65% |
| 2018 | 152 | 1.85x | 1.65x | 100% | 88% |
| 2019 | 152 | 1.88x | 1.67x | 100% | 89% |
| 2020 | 152 | 1.49x | 1.42x | 50% | 61% |

Top-decile picks beat the category average on realised mean WM in **8 of 8** annual decision dates. On the goal-achievement measure the record is more mixed: top-decile picks beat the category in **5 of 8** years, tied in 1 (2014, a weak market period where neither group cleared the hurdle), and trailed the category in 2 (2015 and 2020 — both cohorts where the very top-ranked funds happened to be more concentrated/aggressive names that lagged a broader recovery in their specific 5-year window). This is disclosed rather than smoothed over — see Section 4.

## 3. Pooled Evidence Across All 100 Decision Dates

Because monthly decision dates overlap in the funds and periods they cover, the 8 annual snapshots above are the clearest single view. Pooling all 100 monthly cutoffs (13,574 fund-period observations) gives a larger-sample cross-check of the same pattern.

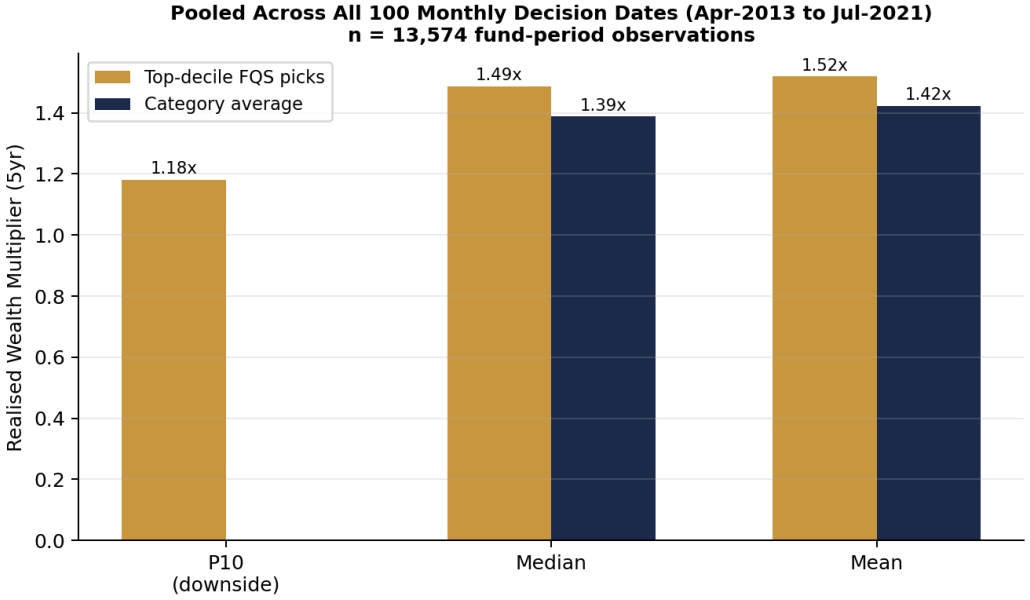



Fig. 3 — Downside (P10), median and mean realised WM: top-decile picks vs category average, pooled.

| Metric | Top-decile FQS picks | Full eligible universe | Difference |
|---|---|---|---|
| **Mean realised WM (5y)** | 1.519x | 1.422x | +0.097x (+6.8%) |
| **Median realised WM (5y)** | 1.487x | 1.388x | +0.099x |
| **P10 realised WM (downside case)** | 1.181x | 1.147x* | +0.034x |
| **Goal achievement rate** | 60.3% | 52.0% | +8.3 pp |

*Universe-wide P10 shown for reference; not a per-cohort figure comparable on a like-for-like basis.

### By fund category

The uplift is not uniform across categories. Mid Cap, Small Cap and Large Cap top-decile picks clearly beat their category average; ELSS, Flexi Cap, Multi Cap and Hybrid top-decile picks actually realised a lower mean WM than their category average over this sample. This is a genuine finding, not noise smoothed away — see Section 4.

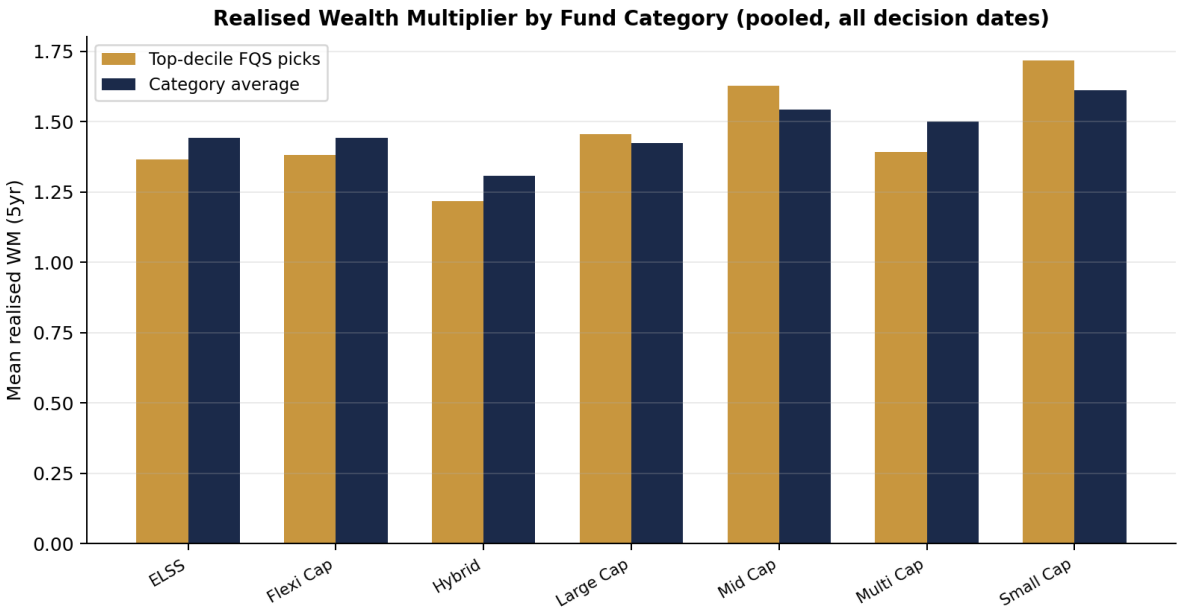



Fig. 4 — Mean realised 5-year WM by category, top-decile picks vs category average, pooled across all decision dates.

## 4. What This Result Does and Does Not Prove

Reporting only the flattering numbers would misrepresent the model's actual, current statistical strength. Two caveats materially qualify the headline result and are carried over directly from the engine's own cross-validation run (`backtest_data/results/accuracy_summary.json`):

**Overlapping windows.** The 100 monthly decision dates are not independent trials — each 5-year forward window overlaps its neighbours by up to 59 months. Statistically, this sample of 13,574 observations behaves like only **~2.65 independent** 5-year periods since 2013. Every rank-correlation figure the engine computed (Spearman ρ = 0.24 for blended_wm, t = 3.34) is explicitly flagged by the engine itself as "not interpretable — too few independent periods." The year-by-year and pooled comparisons above are real, computed outcomes — but they should be read as descriptive evidence over a short live-market history, not as a statistically significant proof the model beats chance.

**Not uniform across time or category.** 2015 and 2020 decision dates, and the ELSS / Flexi Cap / Multi Cap / Hybrid categories, did not show the same uplift — in some cases top-decile picks trailed the category average. A goal-based model should be judged on whether it raises the odds of hitting a target, not guaranteed to win in every period or every category.

**No market-index benchmark.** The "goal-hurdle WM" used above is ArthVruksh's own internal category target (from the FQS scoring table), not a market index like Nifty 50 / Nifty 500 TRI. Comparing against an actual benchmark index is the most useful next step (see Section 5).

**Downside case (P10) undershoots slightly.** The model's stated P10 downside case is meant to hold ~90% of the time; in the full engine run it held 84.8% of the time — modestly miscalibrated, worth tightening.

## 5. Recommended Next Steps

1. Extend the NAV history window (currently constrained to 2006 onward for most funds) so more independent, non-overlapping 5-year periods can accumulate — true statistical significance needs roughly 8–10 independent windows, which requires several more decades of live data or a shorter validation horizon (e.g. 3-year) run in parallel.

2. Add a market-index benchmark (Nifty 500 TRI or a category-matched index) as a second comparison column alongside the category average, so "beat the market" and "beat the peer group" can be reported separately.

3. Re-run this same protocol at a 3-year horizon, where more independent periods are available, as a faster-feedback cross-check on the 5-year conclusions.

4. Investigate the 2015/2020 and ELSS/Hybrid category shortfalls specifically — understand whether they reflect a structural weakness in how blended_wm ranks those fund types, or a genuinely hard market regime for any ranking method.

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All figures in this report are computed directly from ArthVruksh's existing point-in-time backtest engine output (`backtest_data/results/`, full run completed 07-Aug-2026: 164 funds, 100 validated monthly cutoffs, 13,574 fund-period observations). No figure in this report uses NAV data that postdates the relevant decision date when computing a prediction; future NAVs are used exclusively to compute realised outcomes after the ranking was frozen.